

Introduction

Sequence alignment is a key bioinformatics process for comparing the structural, functional, and evolutionary information between two genomes. Sequence alignment involves making the sequences the same length and aligning them, so they have as many matching pairs as possible, by adding gaps into the sequences (Zhang et al., 2022). We have already discovered algorithms that perfectly solve pairwise sequence alignment, when we only have two genomes that we want to compare, but trying to compare three or more sequences is a much harder problem. This is known as multiple sequence alignment, or MSA, and as we want to study longer and more sequences at once it becomes more important to be able to create accurate alignments (Zhang et al., 2022).

Zhang et al. describes five different types of categories that algorithms for multiple sequence alignment can fall into: progressive alignment algorithms, iterative algorithms, heuristics, machine learning, and divide-and-conquer (2022). Progressive alignment algorithms are straightforward, fast, and methods such as ClustalW have been used for multiple sequence alignment for decades. However, progressive alignment algorithms do not always have great accuracy and are easily affected by the number of sequences you want to compare. This led to the development of iterative algorithms that produce better alignment results and are less sensitive to the number of sequences. Iterative algorithms like MUSCLE “continuously optimize the alignment to improve the accuracy until it converges” but this comes with additional time requirements (Zhang et al., 2022). Heuristic algorithms are a type of iterative algorithm that include a factor of randomness and “calculate each feasible solution of the combinatorial optimization problem within the acceptable range of time or space” (Zhang et al., 2022). There are many different types of heuristics algorithms, but because they all feature randomness, the results may be different each time (Zhang et al., 2022). Applying machine learning to multiple sequence alignment is still in its initial stages but shows some promise. Machine learning usually uses a process similar to progressive alignment, but a lack of good alignment samples makes development difficult (Zhang et al., 2022). Divide and conquer algorithms break down problems into smaller problems for processing and is a standard solution for solving multiple sequence alignment. Divide and conquer algorithms like FMAAlign and others have a lot more flexibility than other algorithms because of how they break problems down, while also increasing efficiency (Zhang et al., 2022).

Genetics has slowly been moving toward wanting to compare more and more sequences at once. Zhang et al. gives the example of over ten thousand genomes being collected to analyze the variation in SARS-CoV-2 (Zhang et al., 2022). These hyper scale sequences are incredibly challenging to deal with using current multiple sequence

alignment techniques. Current MSA techniques also find it difficult “to predict the evolution process of sequences correctly and lack explanations” (Zhang et al., 2022). This may be due to the random nature of evolution, the limitations of the technique, or the lack of an accurate evolutionary model. Multiple sequence alignment is a foundational bioinformatics technique, but there are still strides to be made in developing better algorithms.

Methods

Even after addressing the challenge of creating a multiple sequence alignment using any of the techniques mentioned previously, there is a challenge in visualizing the relationship between the sequences. Here, I used Python to try to visualize a multiple sequence alignment in two different ways. I first used MUSCLE to perform the multiple sequence alignment due to it being assessable and fast. I then used the pyMSAvis package to produce a chart with each nucleotide colored different. I then used the network module in Python to produce a flow chart displaying the differences between the three sequences.

Results

Using the pyMSAvis package in Python, I produced Figure 1. This shows the alignment of the three provided sequences after performing a MSA using MUSCLE. I then used the alignment to produce a diagram more clearly showing the similarities and differences between the three genomes, shown in Figure 2. This shows only the first one hundred base pairs of the given genomes.

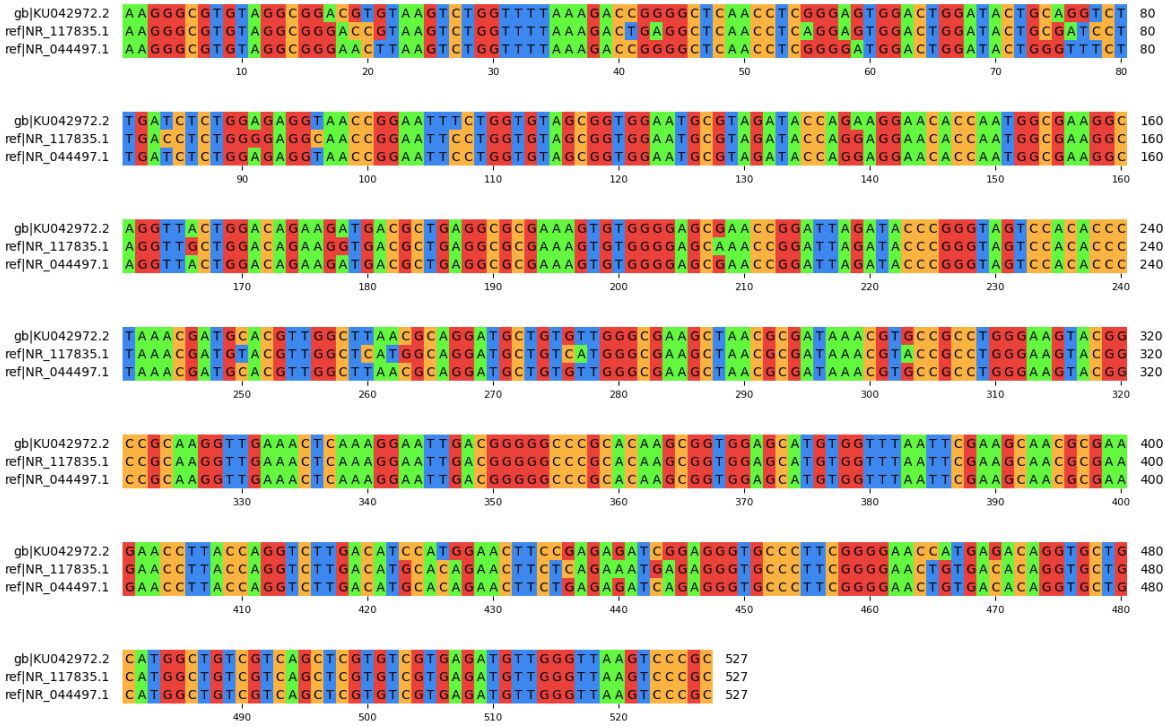


Figure 1. Multiple sequence alignment chart for three given sequences.

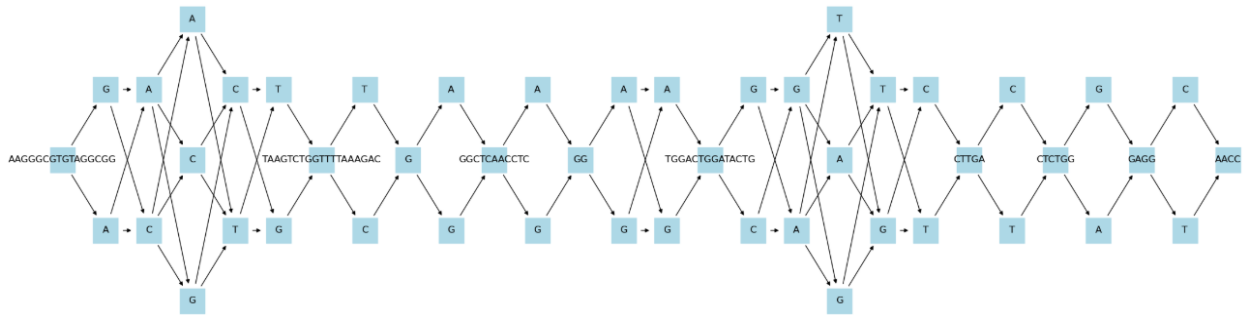


Figure 2. Multiple sequence alignment flow chart for the sequences shown in Figure 1.

Conclusions

Visualizing a multiple sequence alignment in a clean graph is a challenging task, as there is so much data you must make decisions on what is most important for you to show. Figure 1 clearly shows how much similarity there is between the sequences, and it is easy to understand, but the different colors and vast amounts of characters make it harder to really pick out the differences, because they can blend in. I think a chart like Figure 1 can be helpful after performing an alignment to see at a quick glance how good the alignment

is, but it lacks a lot of value as a visualization after that. Figure 1 also would not scale well. Here we are using only three short sequences, but trying to produce a similar chart with a lot more sequences or much longer sequences would make it much harder to read and much less useful as a tool to compare genomes.

Figure 2 makes the differences between the genomes much more obvious, but the vast similarities become harder to see. I will admit that on my part, Figure 2 is not a great visualization due to my own lack of familiarity with the networkx Python package. Some of the faults might lie with my visualization in particular, so I will generalize my comments to similar graphs like those produced by GenGraph in Figure 3 of Ambler et al. (2019). Having the difference base pairs in different boxes when there is a difference between the genomes, even if it is only one base pair different, makes these differences really stick out as compared to Figure 1 where the differences blend in. The similarities, though much longer in this case, get less representation on the graph. A flow chart like this could be scaled for longer sequences without becoming too busy, but it would depend on how similar the sequences were. Trying to show more than three sequences in an alignment would start creating a very messy graph as well.

I do not think either one of these visualizations is enough on its own to show the similarities and differences between the sequences in a nice way, but together they help paint a picture of what the alignment of these sequences looks like. We see that most of the sequences are very similar, but in the first one hundred base pairs there are differences at 18 base pairs, so approximately 18% is different. I think that which visualization method you would want to use would depend on your use case for it, whether you want to highlight the differences or the similarities between the sequences. However, both tools are limited in their scope and work best with short sequences and only 2 or 3 sequences. I think they could be a valuable tool for comparing a genome to a reference genome but have less application within multiple sequence alignment as MSA shifts towards comparing more sequences and longer sequences.

References

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